

Received 9 December 2024, accepted 6 February 2025, date of publication 13 February 2025, date of current version 20 February 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3541878

RESEARCH ARTICLE

A Novel Weighted Loss TabTransformer Integrating Explainable AI for Imbalanced Credit Risk Datasets

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This work was supported in part by the Vice-Rector of Research, Innovation, and Entrepreneurship at Satya Wacana Christian University through the Article Writing Bootcamp Program and Internal Research Grant.

ABSTRACT Credit risk assessment often faces significant challenges due to class imbalance and the opaque nature of machine learning models, which can result in biased predictions and hinder trust among stakeholders. To address these issues, this study proposes a framework combining the TabTransformer model with weighted loss techniques to balance class distributions and improve predictive accuracy. Applied to the BISAID and German Credit datasets, the method demonstrated notable improvements in accuracy, from 86.35% to 89.27% and 93% to 95%, respectively, along with improved minority class AUC and precision-recall metrics. To ensure transparency and interpretability, SHAP (SHapley Additive exPlanations) was employed, highlighting critical predictors such as “Financing Needs” and “Credit Amount.” By integrating fairness mechanisms through weighted loss and explainability via XAI, the proposed framework and weighted loss TabTransformer mitigate bias, enhance model performance, and provide actionable insights for borrowers and stakeholders. These findings establish a reliable, equitable, and transparent approach to credit evaluation.

INDEX TERMS Class imbalance, credit risk assessment, deep learning, classification, explainable artificial intelligence.

I. INTRODUCTION

Credit risk assessment is a critical process that applies to all borrowers, ranging from individuals to businesses, including micro, small, and medium enterprises (MSMEs). In Indonesia, MSMEs form a strategic sector that supports the national economy, comprising 65.46 million units, which is significantly higher than Thailand’s 3.1 million and Malaysia’s 1.2 million units [1], [2]. This sector contributes 60.3% of Indonesia’s Gross Domestic Product (GDP), employs 97% of the workforce, and accounts for 14.4% of national exports [3]. Despite their economic importance, MSMEs often face obstacles in accessing formal business financing through commercial or rural banks [4], [5]. Weak access to formal funding drives many MSMEs to rely on informal funds,

which impose higher interest rates and reduce competitiveness and profitability [6]. Beyond MSMEs, access to credit is a critical issue for many types of borrowers in Indonesia, reflecting broader challenges in financial inclusivity. In this context, cooperatives and innovative financial technologies play a vital role in bridging the gap, providing affordable and accessible financing options for both individual and business borrowers [7].

The problem of limited financial access for borrowers is rooted in the issue of information asymmetry, which arises due to the unequal distribution of information between borrowers and financial institutions. This asymmetry arises from unequal information distribution, allowing borrowers to obscure or manipulate information (adverse selection) or misuse credit (moral hazard) [8], [9]. In developing countries, the risk of financial information asymmetry is more prevalent compared to developed nations [10]. A significant

The associate editor coordinating the review of this manuscript and approving it for publication was Sawyer Duane Campbell¹.

factor contributing to this asymmetry is the lack of audited financial statements from borrowers, which makes it challenging for formal financial institutions to evaluate their creditworthiness. The Bank of Indonesia has developed a database profiling MSMEs, and the Statlog German credit approval dataset (containing 1,000 instances) is available in the UCI Machine Learning Repository [11]. These resources aim to assist financial institutions in reducing information asymmetry. However, challenges persist in effectively analyzing and utilizing this data [12].

With the use of various tools to address the problem, Machine Learning methods have been of particular importance. Some studies indicate that lending analytics using ML techniques is possible [13], [14]. However, there is a significant limitation in many conventional ML approaches: they are not interpretable; people tend to refer to them as ‘black boxes’ which make predictions but do not illustrate how the prediction was made [15]. This lack of thinking may create a situation where trust and responsibility in the automatic evaluation of a borrower’s creditworthiness becomes infeasible. Some ML techniques also have some problems such as overfitting [16], where models developed in training data are unable to generalize on unseen data [17]. Further complicating the situation is the fact that all the financial institutions have their unique policies and requirements for deciding the eligibility of a loan thus making it very difficult to come up with a standard model. These problems create an urge for new models that will not only improve predictive performance but also enable institutions to have interpretable and flexible models that suit different financial institutions.

Deep Learning presents a solution to the challenges faced in assessing creditworthiness due to its ability to detect patterns and trends in structured data, such as that provided by BISAID and the German credit, as proven by other research that DL methods often outperform traditional statistical and machine learning approaches [18], [19], [20], [21], [22], [23], [24]. It is essential to consider all relevant factors influencing credit decisions, including unexpected ones [25], [26]. The evolving paradigm of Deep Learning, Explainable Artificial Intelligence (XAI), has shown significant advancements. XAI enables the explanation of influential factors in a detailed and human-readable manner, providing insights into the rationale behind credit decisions [27], [28]. This capability enhances the quality of credit evaluations by formal financial institutions, making the MSME credit decision process more efficient and effective. Moreover, XAI highlights factors that conventional AI approaches may overlook, which typically provide loan recommendations without clarifying the decision-making factors involved.

Despite these advancements, one of the critical challenges in applying deep learning and XAI to MSME credit assessment lies in addressing data imbalance. In many cases, instances of credit default are significantly less frequent than successful credit outcomes, creating a skewed dataset that can bias the model’s predictions toward the majority class [29].

To counter this, the use of weighted loss during model training assigns greater importance to minority classes, such as defaults, allowing the model to better capture high-risk patterns that might otherwise be overlooked. This approach not only improves the model’s accuracy but also ensures that XAI outputs remain reliable and unbiased. When minority classes are underrepresented, XAI interpretations can become skewed, highlighting trends in the majority class while under-emphasizing critical risk indicators present in the minority. Therefore, integrating weighted loss with XAI enhances both predictive accuracy and interpretability, providing financial institutions with balanced, transparent insights that are essential for fair and effective MSME credit evaluations.

The main research problem centers on the difficulty financial institutions face in evaluating the creditworthiness due to significant information asymmetry. Despite the availability of structured data from platforms like BISAID, many MSMEs lack audited financial statements, making it challenging for lenders to accurately assess credit risk. This asymmetry—where lenders and borrowers possess unequal knowledge about financial health—creates higher perceived risks for financial institutions, hindering their ability to provide adequate financing for borrowers and limiting their potential growth. In addition, financial institutions struggle to determine the appropriate weighting of financial data during the credit decision-making process, leading to inefficiency and inconsistency in assessments. Conventional AI methods further complicate this issue by offering limited transparency, which can erode trust among MSMEs regarding loan decisions. To address these issues, this research proposes a deep learning-based model that utilizes BISAID’s dataset and German Credit dataset to more effectively assess creditworthiness. The model will incorporate Explainable Artificial Intelligence (XAI) techniques to clarify the factors influencing loan decisions, providing transparency for both financial institutions and borrowers. By establishing a clear approach to weighing financial indicators, the proposed method aims to enhance the efficiency, transparency, and fairness of credit assessments, fostering greater financial inclusivity for all types of borrowers.

II. RELATED WORKS

A. CLASSIFICATION APPROACHES IN DEEP LEARNING

The development of classification methods in Deep Learning has evolved from traditional Neural Networks (NN) to more sophisticated architectures aimed at addressing specific challenges, such as processing sequential data and improving training efficiency [30]. Early classification tasks primarily relied on simple feed-forward Neural Networks, which consist of fully connected layers designed to capture patterns in input data through multiple hidden layers. While these networks were effective in many applications, they struggled with time-series or sequential data due to their inability to retain the memory of past inputs [31], [32].

To overcome this limitation, Recurrent Neural Networks (RNNs) were introduced [33], which allowed the model

to retain information from previous inputs through hidden states. However, standard RNNs suffered from vanishing gradient problems, which hindered their ability to capture long-term dependencies. The Gated Recurrent Unit (GRU), a simplified version of the LSTM, was developed to mitigate this issue by introducing gating mechanisms that control the flow of information [34]. With the rise of e-commerce platforms for financial transactions, fraudsters also exploit these systems. Fraud prevention systems (FPSs) are often inadequate. The proposed method involves preprocessing, feature selection, and model training. Preprocessing uses discretization and min-max normalization, while GA-based feature selection enhances model efficiency. A BiGRU-A-CapsNet model was applied, outperforming standard BiGRU and CapsNet, achieving 95.44% accuracy, and making it suitable for intrusion detection and fraud prevention [35].

The Transformer architecture, introduced by [36], addressed these limitations by replacing recurrence with self-attention mechanisms, allowing the model to process entire sequences simultaneously. Transformers demonstrated superior performance in tasks requiring global context, such as NLP, and also proved to be more parallelizable, significantly speeding up training times. This architecture has since become a standard in deep learning for various classification tasks, such as Autoformer [37], which uses Autocorrelation to enhance long-term forecasting, Vision Transformer (ViT) [38], and HSI-BERT [39] for image classification.

In recent years, the TabTransformer [40] has emerged as an adaptation of the Transformer architecture specifically designed for tabular data, which often consists of a mix of categorical and numerical features. Extensive experiments on fifteen publicly available datasets show that TabTransformer outperforms state-of-the-art deep learning methods for tabular data by at least 1.0% on mean AUC, while matching the performance of tree-based ensemble models. Additionally, it is highly robust against missing and noisy data, offering improved interpretability through its learned contextual embeddings. In semi-supervised settings, an unsupervised pre-training procedure further enhances performance, leading to an average 2.1% AUC improvement.

B. WEIGHTED CROSS-ENTROPY LOSS

Weighted loss serves as a well-regarded approach to handle disproportion among categories within machine learning duties, specifically in categorization difficulties where the distribution of groups is uneven. Such a lack of balance, which arises when certain groupings are underrepresented, can guide model prejudice where the classifier favors more recurrent groupings, eventually reducing execution on minority clusters [41]. To counteract this, weighted penalty functions allocate higher punishments to scant groupings, consequently stabilizing the effect of each grouping on the model's studying method. Weighted penalty functions have been applied efficiently in domains for instance healthcare imagery [42] and agriculture [41], [43], where class distributions are often imbalanced owing to the unusualness

of definite problems or illnesses. Furthermore, the approach holds promise in dealing with natural language data where common words contribute disproportionately to training and rarer terms are more challenging to learn accurately. Thus, weighted loss helps achieve a richer overall model that more evenly represents the full spectrum of language.

In two studies investigating the application of weighted loss on imbalanced datasets, namely for brain tumor and maize disease classification, each demonstrated significant improvements in model performance metrics. The by Lundberg and Lee [44] showed that deep feature fusion and weighted cross-entropy led to increases in accuracy, precision, and F-score by over 10% compared to standard cross-entropy, with gains in minority classes. Furthermore, the study by Ahadian et al. [29] employed transfer learning alongside weighted loss to achieve nearly 90% accuracy, 88% precision, and a substantial increase in F-score, illustrating the effectiveness of weighted loss in mitigating class imbalance.

The weight for each class w_i in weighted cross-entropy loss is determined by:

$$w_i = \frac{N}{N_i} \quad (1)$$

where N represents the total samples and N_i the samples in class i , ensuring higher weights for minority classes. The weighted cross-entropy loss $L_{\text{Weighted Cross-Entropy Loss}}$ is then defined as:

$$L_{\text{Weighted Cross-Entropy Loss}} = - \sum_{i=1}^C w_i \cdot y_i \cdot \log(\hat{y}_i) \quad (2)$$

with C as class count, y_i as true labels, and $(\hat{y}_i)$ as predicted probabilities.

C. EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) FOR CLASSIFICATION TASKS

Explainable AI is a subdomain of artificial intelligence. It deals with the development of techniques that provide more interpretable and understandable machine learning models, especially those using deep learning approaches [41]. While these models have continued to increase in complexity, their decision-making processes often resemble “black boxes” and make explanations of the predictive results quite difficult. XAI addresses this issue by providing insights into which features influenced a model's decision, offering much-needed transparency in high-stakes fields like healthcare, finance, and legal applications [42]. Various approaches have been developed to provide explanations for classification tasks. A representative method is LIME, which explains individual predictions by approximating complex systems with simpler, more interpretable models, such as linear classifiers, acting locally around the instance in question [43]. On the other hand, SHAP is a powerful method based on game theory, which assigns a Shapley value to each feature for quantify its contribution to a prediction [44]. SHAP is frequently used in the context of tabular data classification, providing

detailed insights into how individual features contribute to a model's prediction by assigning each feature a Shapley value, ensuring transparency and interpretability [45].

The XAI technique, Anchors, explains by identifying specific conditions or rules that invariably lead to similar predictions, thus providing a high degree of precision, explain by identifying specific conditions or rules that invariably lead to similar predictions, thus providing a high degree of precision [43]. An alternative methodology involves counterfactual explanations, which emphasize the provision of substitute inputs that would lead to divergent outcomes, thereby illustrating how minor modifications in the input can influence the prediction [46]. Methods that assess feature importance are frequently applied within tree-based models, such as Random Forests, allocating scores to each input feature according to their impact on the accuracy of the model's predictions [47]. Similarly, saliency maps, originally designed for image processing tasks, compute the gradient of the output with respect to each input feature, highlighting which parts of the input most influence the model's decision [48].

Most of the approaches in XAI generally fall into two distinct categories: model-agnostic and model-specific. Examples are LIME and SHAP, model-agnostic approaches that are versatile because they apply to any type of machine learning model. Model-agnostic approaches, such as Grad-CAM and feature importance associated with decision trees, are inherently tied to specific model architectures and provide insights specific to their inner workings. In contrast, model-agnostic techniques such as SHAP are often preferred because they generalize across a wide array of models in a consistent manner, enabling consistent explanations regardless of the underlying model architecture. SHAP stands in a unique position because the theoretical foundation behind it in cooperative game theory enables fair and reproducible feature attribution, ensuring that all feature contributions collectively equal the model's predicted output. Therefore, SHAP can be an effective tool to explain individual predictions and understand the overall behavior of models. Various studies have highlighted the effectiveness of SHAP in tasks like tabular data classification. For example, [49] used SHAP to explain tree ensemble models in healthcare, providing insights into how patient features impacted diagnosis predictions. Reference [50] applied SHAP to cybersecurity anomaly detection, identifying the key features driving malicious activity detection. Similarly, [51] demonstrated SHAP's value in finance, explaining credit risk predictions by highlighting important customer features like income and credit history. In telecommunications, [52] used SHAP for churn prediction, identifying factors such as service usage and contract duration that influence customer retention. Lastly, [53] employed SHAP in fraud detection within banking, clarifying how transaction patterns and account history contribute to flagging suspicious activities. These examples illustrate SHAP's versatility and effectiveness in delivering actionable insights across various classification tasks.

D. The SHAP value for a feature i is calculated using the following formula:

The SHAP value for a feature i is calculated using the following formula:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! \cdot (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \quad (3)$$

where:

- N is the set of all features.
- S is a subset of features not containing feature i .
- $f(S)$ is the model's prediction for the set of features S .
- ϕ_i is the Shapley value for feature i , representing its contribution to the model's output.

III. PROPOSED METHOD

Figure 1 illustrates the deep learning pipeline for loan acceptance rate prediction using the BISIAID and German Credit datasets. The process begins with data collection, followed by preprocessing steps such as feature engineering (e.g., data imputation, one-hot encoding), normalization, and data splitting (80% training, 20% testing). The TabTransformer model, enhanced with a weighted loss function to address class imbalance, serves as the core architecture, undergoing hyperparameter tuning and multiple evaluations to ensure accurate predictions. Post-modeling, Explainable AI (XAI) techniques, specifically Shapley values, are applied to interpret predictions, providing local explanations for borrowers to understand the factors influencing their loan outcomes.

A. DATA COLLECTING

The data is retrieved from the BISIAID database, containing various features related to MSME profiles such as SEKTOR (sector), TENAGA_KERJA (number of employees), TOTAL_ASET (total assets), PENJUALAN_TAHUN (annual sales), and KEBUTUHAN_PEMBIAYAAN (financing needs). These features are essential for training the model, which is designed to predict loan acceptance rates based on criteria verified by official Bank of Indonesia. Meanwhile, the German Credit dataset includes attributes like age, job type, housing status, credit amount, and loan duration, focusing on individual credit profiles. Together, these datasets offer a comprehensive foundation for reducing information asymmetry and training predictive models.

B. DATA PREPROCESSING

The Preprocessing Data phase is a crucial step that transforms raw data into a form suitable for machine learning models. This process begins with feature engineering, where various transformations are applied, including data imputation (addressing missing values), one-hot encoding (converting categorical variables into binary format), and discretization (segmentation of continuous variables into discrete intervals). These techniques ensure that the data can be effectively interpreted by the deep learning model.

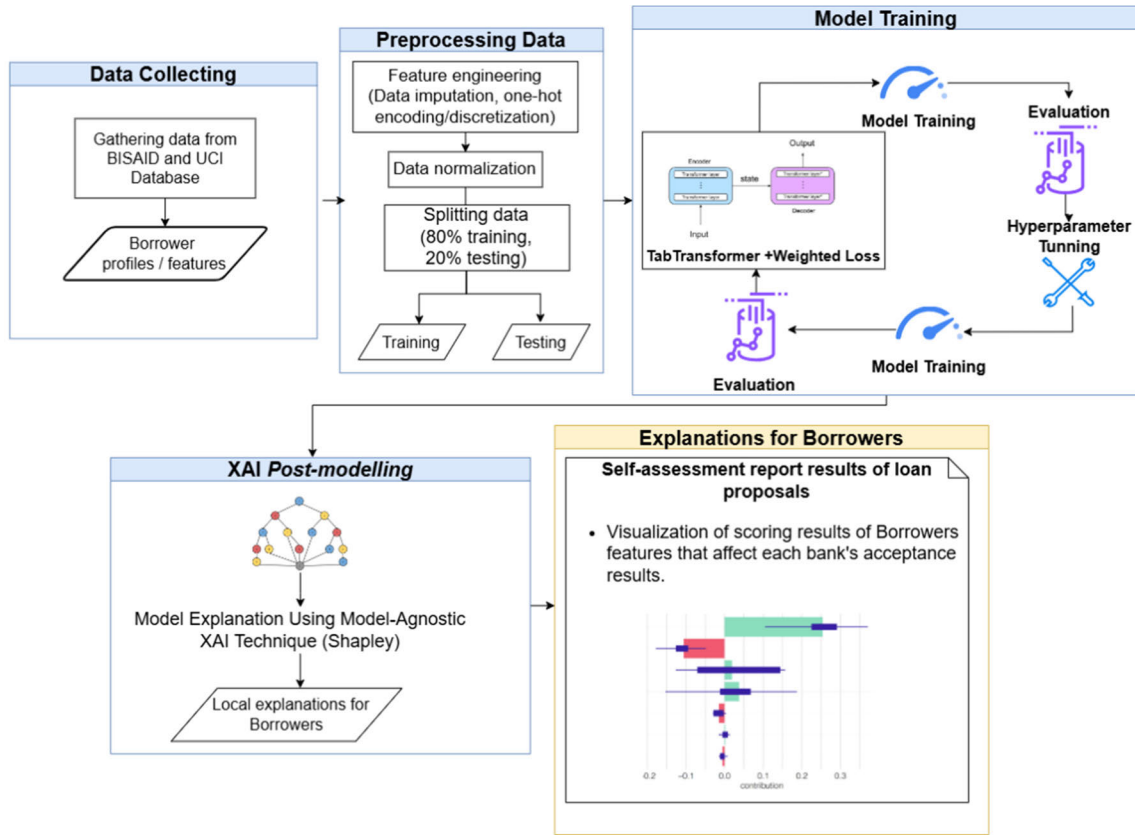


FIGURE 1. Proposed framework utilizing TabTransformer and weighted loss for credit risk prediction and explainability.

Following feature engineering, data normalization is performed to standardize feature ranges, ensuring consistent scaling across features to prevent any single feature from disproportionately influencing the model. In this study, normalization is applied using a range of -1 to 1 , which helps maintain numerical stability and improves model performance. Finally, the dataset is divided into training (80%) and testing (20%) subsets, facilitating the model's development and subsequent evaluation on unseen data to assess its generalizability.

C. MODEL TRAINING

Before initiating the model training process, class weights were calculated as required for the weighted loss function, with the formula for computing these weights provided in Equation 1. The Model Training phase employs the TabTransformer model. This approach offers a more effective solution compared to traditional models, which often struggle with these relationships. Figure 2 demonstrates how TabTransformer works, started by embedding categorical features and passing them through Transformer layers, leveraging multi-head attention to capture complex interactions, while normalized continuous features are concatenated with the Transformer output. This combined representation is processed by a Multi-Layer Perceptron (MLP) for predictions, making TabTransformer a versatile model for handling

complex feature dependencies in tabular datasets where traditional models may struggle.

The evaluation of classification models gives rise to various key metrics that come from the confusion matrix which contains True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). These components are identified in the following sense: True Positives (TP) signs represent cases in which the model accurately identifies positive examples, whereas False Positives (FP) occur when the model wrongly defines a negative sample as positive. In contrast, the True Negatives (TN) signify the model correctly identifying a negative sample, and the False Negatives (FN) stand for the instances where the model wrongly classifies a positive sample as negative. By these basic concepts, key performance metrics are derived to disclose the model's performance in detail.

Accuracy is a measure of the overall correctness of the model and is defined as the proportion of correctly classified instances (both positive and negative) over the total number of instances. The formula is given by:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

Precision, also known as the positive predictive value, measures the proportion of true positive predictions among all positive predictions made by the model. This metric

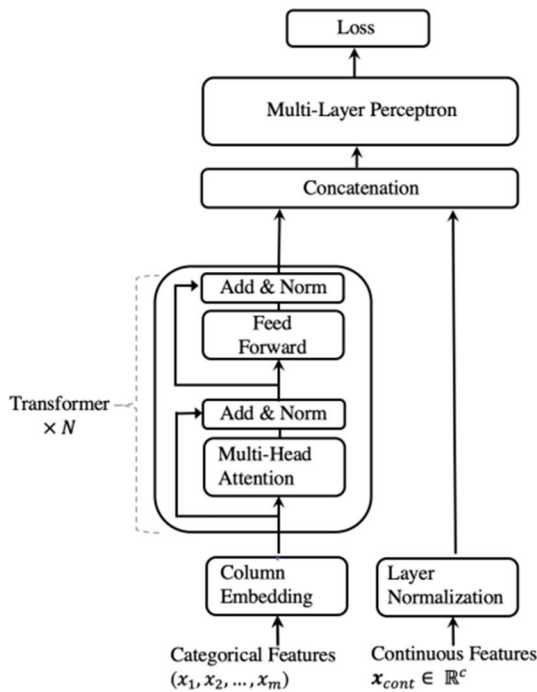


FIGURE 2. TabTransformer architecture by Huang et al. [40].

is particularly important in contexts where the cost of false positives is high. Precision is computed as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

High precision indicates that the model makes few false positive errors, which is crucial in domains such as fraud detection or medical diagnosis, where incorrect positive predictions can be costly.

Recall, also referred to as sensitivity or the true positive rate, quantifies the model's ability to correctly identify all relevant positive cases. It is particularly useful in scenarios where the cost of false negatives is high, as it reflects the proportion of actual positive instances that are correctly identified by the model. Recall is defined as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (6)$$

F2 Score is a variant of the F-measure, which provides a balance between precision and recall. However, unlike the F1 score, the F2 score places greater emphasis on recall, making it more suitable in situations where false negatives are more critical than false positives. The F2 score is calculated as follows:

$$F2 = \frac{5 \times \text{Precision} \times \text{Recall}}{4 \times \text{Precision} + \text{Recall}} \quad (7)$$

These metrics provide a comprehensive evaluation framework for TabTransformer model, helping to assess its effectiveness in handling tabular data classification tasks, particularly with complex feature interactions.

The training process is iterative, incorporating hyperparameter tuning to optimize key parameters such as learning rate, batch size, and attention heads. Multiple rounds of training and testing are conducted to enhance the model's predictive performance. The model's effectiveness will be gradually evaluated using metrics outlined in Equations 4-7.

D. XAI POST-MODELING

Following the model's development, the XAI (Explainable Artificial Intelligence) Post-Modeling phase is implemented to provide transparency and interpretability of the model's predictions. In this phase, model-agnostic techniques, such as Shapley values, are applied to offer both local and global explanations of the model's decision-making process. Shapley values, rooted in cooperative game theory, allocate contributions to individual features based on their influence on the prediction, providing insight into how specific MSME characteristics (e.g., credit score, business tenure) affect loan acceptance probability.

IV. RESULTS AND DISCUSSIONS

A. DATA COLLECTING

Data in this study was collected from an official government website that has been managing this information since its inception. In this study, the researchers obtained 2563 records, each with 12 features. Among these features, five were removed because they contained only identity-related information, such as reference code, research year, and owner name, and so on. The remaining features are a mixture of categorical and numerical data, which is ideal for the TabTransformer architecture, as shown in Table 1.

TABLE 1. BISAID dataset feature types and descriptions.

Variable Name	Variable Type (Numeric/Categorical)
City	Categorical (225 unique values)
Province	Categorical (33 unique values)
Sector	Categorical (19 unique values)
Labour	Numerical
Total Assets	Numerical
Annual Sales	Numerical
Financing Needs	Numerical

In the German Credit dataset, a total of 1,000 instances were obtained, each containing twice as many features per instance compared to the features in other datasets. These features were also compared with data from BISAID, which demonstrated fewer variations in feature categories. Specifically, only the "purpose" feature exhibited 10 unique values, while the remaining features ranged from 5, 4, 3, to 2 unique values, as presented in Table 2.

B. DATA PREPROCESSING

The primary objective of this research is to examine the features identified by the TabTransformer model. Before the data is processed by the model, it is compulsory to ensure that

TABLE 2. German credit dataset feature types and descriptions.

Variable Name	Variable Type (Numeric/Categorical)
Telephone; Foreign worker	Categorical (2 unique values)
Guarantors; Other installment plans; Housing	Categorical (3 unique values)
Status of existing checking; Property; Job	Categorical (4 unique values)
Credit history; Savings account; Present employment since; Personal status and sex; Purpose	Categorical (5 unique values)
Duration (months); credit amount; Installment rate; Present residence since; Age (years); Number of existing credits; Number of liable people to provide maintenance	Categorical (10 unique values)
	Numerical

the dataset is consistent, particularly in terms of numerical data. To minimize bias across the training, validation, and test sets, class distributions must be balanced. Accordingly, this study employs a data split ratio of 80:10:10 for training, validation, and testing, respectively. Figure 3 depicts the partition of the train, validation, and test sets, showing the balanced distribution of classes. Each bar in the figure is segmented to represent the proportion of each class (0, 1, 2, and 3) within the respective datasets. The colors—blue for class 0, green for class 1, yellow for class 2, and red for class 3—clearly illustrate the class distribution across all subsets, ensuring no bias is introduced during the data splitting process.

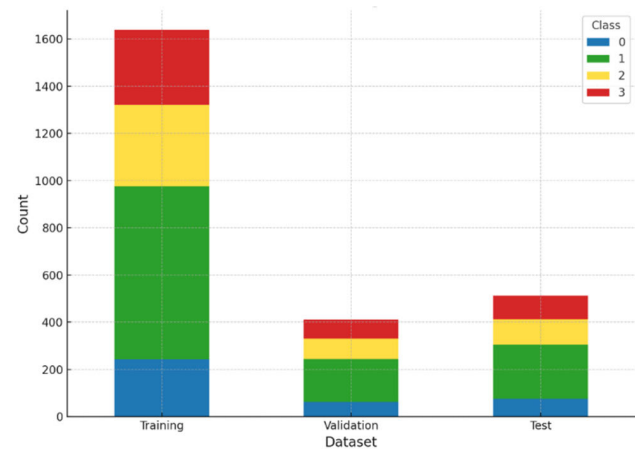


FIGURE 3. BISAID class distribution across dataset splits.

The German Credit dataset, obtained from the UCI Machine Learning Repository, consists of two classes: Class 0 and Class 1. As shown in Figure 4, Class 0 represents “bad customers” with 300 instances, while Class 1 represents “good customers” with 700 instances. To maintain consistency across the training, validation, and test datasets, the same 80:10:10 data split ratio as used in BISAID was applied, as shown in Figure 3. The distribution within each set was ensured to be consistent, preventing bias toward the

majority class and supporting fair evaluation of the weighted loss performance during inference with XAI.

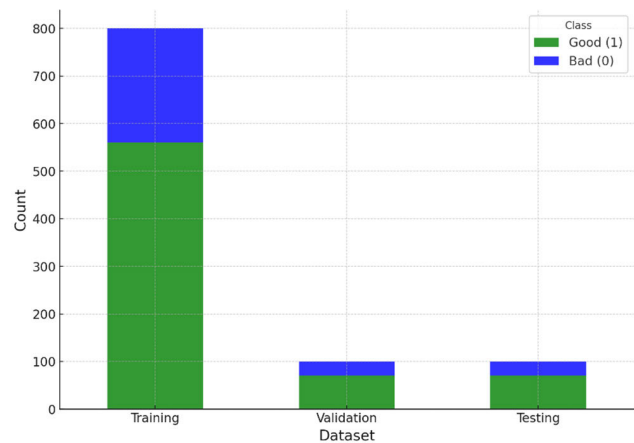


FIGURE 4. German Credit class distribution across dataset splits.

C. MODEL TRAINING

Before training the model, the Weighted Class Loss must be configured first. In this study, the author uses the original dataset distribution prior to splitting it into various subsets, as shown in the first figure. In the dataset containing 2563 samples, the distribution of classes from 0 to 3 is 379, 1146, 539, and 499, respectively. Table 3 shows the results of the calculation based on Equation (1), where Class 1 has the smallest penalty (2.2364) due to being the majority class. In contrast, the minority class, Class 0, has the highest weight (6.7625), as the model is rarely exposed to this class. The same principle applies to the German Credit Dataset, where Class 1 (Good) has a lower weight compared to Class 0 because Class 0 is the minority. Therefore, the penalty is higher, as described in the formula, ensuring the model pays more attention to this class. This weighting scheme will later be applied to the loss function, as detailed in Equation (2).

TABLE 3. Weighted loss calculation for each dataset’s classes.

Class	BISAID Dataset Weight	German Credit Dataset Weight
0	6.7625	3.33
1	2.2364	1.43
2	4.7551	-
3	5.1362	-

After class weighting using Weighted Class Loss, the next step is hyperparameter tuning on the TabTransformer, considering accuracy, precision, and recall as the main metrics, evaluated using 10-fold cross-validation, as shown in Table 3. The selected for BISAID dataset hyperparameters are dim = 128, dim_out = 4, depth = 1, heads = 2, attn_dropout = 0.1, and ff_dropout = 0.1. The feature dimension (dim) was increased to 128 from the paper recommendation (32) to capture more complex patterns as the data

contains more categorical unique values, while the output dimension (dim_out) was adjusted to the number of classes 4 (Multiclass) for BISAID dataset and 2 (Binary) for German Credit dataset. The depth was set to 1, lower than the recommendation (6), and the number of attention heads was reduced to 2 from 8 to reduce computational complexity.

The dropout for the attention mechanism (attn_dropout) and feed-forward layer (ff_dropout) were set to 0.1 each to prevent overfitting without reducing the model capacity. These hyperparameters were chosen based on experiments showing that this combination provides the best balance between performance and efficiency.

TABLE 4. TabTransformer hyperparameter settings.

Hyperparameter	BISAID Dataset	German Credit Dataset
Dim	128	64
Dim_out	4	2
Depth	1	2
Heads	2	2
Attn_dropout	0.1	0.1
Ff_dropout	0.1	0.1

After completing the class weight initialization and selecting the best hyperparameters during the tuning phase, the next step is to train the model. In this phase, Table 2 is consistently referenced in Equation (2), as are the hyperparameters in Table 4. During the training process researchers used mini-batch with batch size 64 as recommended study by Ioffe [54]. Figure 5 presents a comparison of the loss between the training data and the validation data. Additionally, the evaluation at each epoch shows consistent fluctuations, indicating that the model struggles to manage the effects of the imbalanced class distribution during training. On the other hand, Figure 1 B shows a more stable train loss, even though it started with a higher initial loss and experienced a few spikes between epochs 0 and 40. Beyond this range, the loss remained stable due to the implementation of weighted loss, which helped address the class imbalance effectively.

Similarly, in the German Credit dataset, a similar pattern is observed in Figure 5. In Figure 5 (a), the model behaves differently compared to the BISAID dataset. Both the training and validation loss curves are spiky, with the model showing signs of near-overfitting across more epochs. The rapid decline in validation loss indicates that, if training is continued, overfitting is highly likely. In contrast, Figure 6 (b) demonstrates the effect of weighted loss, which smooths both the training and validation curves. Additionally, it shows a similar behavior where the initial loss values are generally higher compared to training without weighted loss.

After training the model using weighted loss, the evaluation phase begins, leveraging the previously trained data. In this phase, a classification report as can be seen in Table 5 is generated for each class, providing accuracy, precision, and recall metrics. This approach ensures a fair assessment of the effectiveness of the weighted loss method.

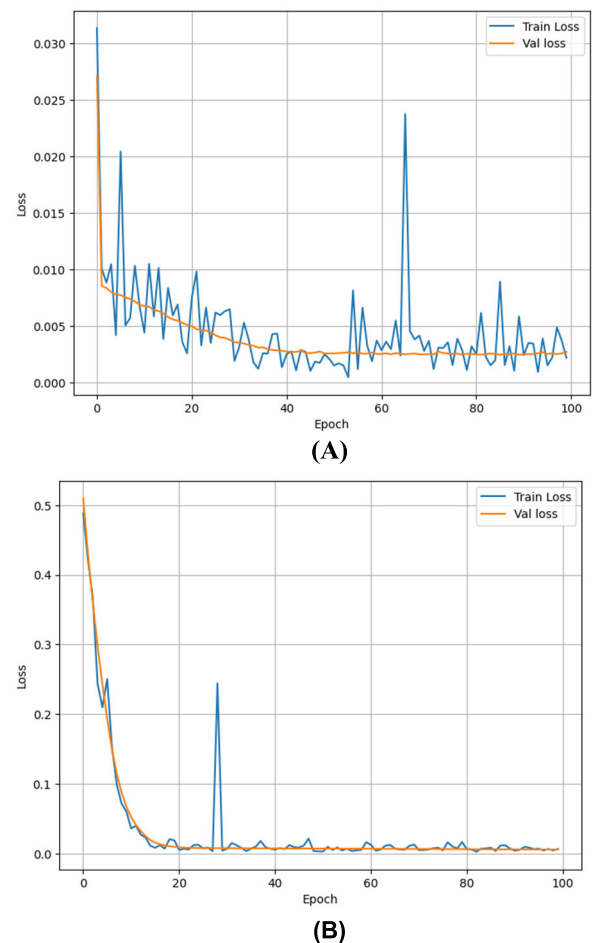


FIGURE 5. Train vs Val loss for BISAID dataset (A) Before weighted (B) After weighted loss.

Beyond addressing data imbalance, the evaluation also tests the model with different features. Furthermore, the study incorporates both binary (two-class) and multi-class classifications to analyze the impact of weighted loss comprehensively. The table demonstrates the improvements brought by the weighted loss. Notably, class 0 shows an increase in precision, albeit with a slight dip in recall. Class 1 and 3 exhibit consistent gains across all metrics, reflecting the positive impact of the weighted loss on performance. However, class 2 sees a trade-off with a slight drop in precision but a significant gain in recall, improving its overall F1-score. The increase in overall accuracy from 86% to 89% further underscores the effectiveness of the weighted loss in enhancing the model's performance to keep aware about minority classes, particularly in handling multi class imbalance.

Figure 7 presents the additional evaluation results using the ROC (Receiver Operating Characteristic) Curve for the four classes. The minority classes (2 and 3) show noticeable improvement in AUC, increasing from 0.88 to 0.91 and 0.86 to 0.91, respectively. For the remaining classes (0 and 1), Class 0 shows minimal change, with weighted and non-weighted models differing by only 0.1, where the non-weighted model slightly outperforms. The majority

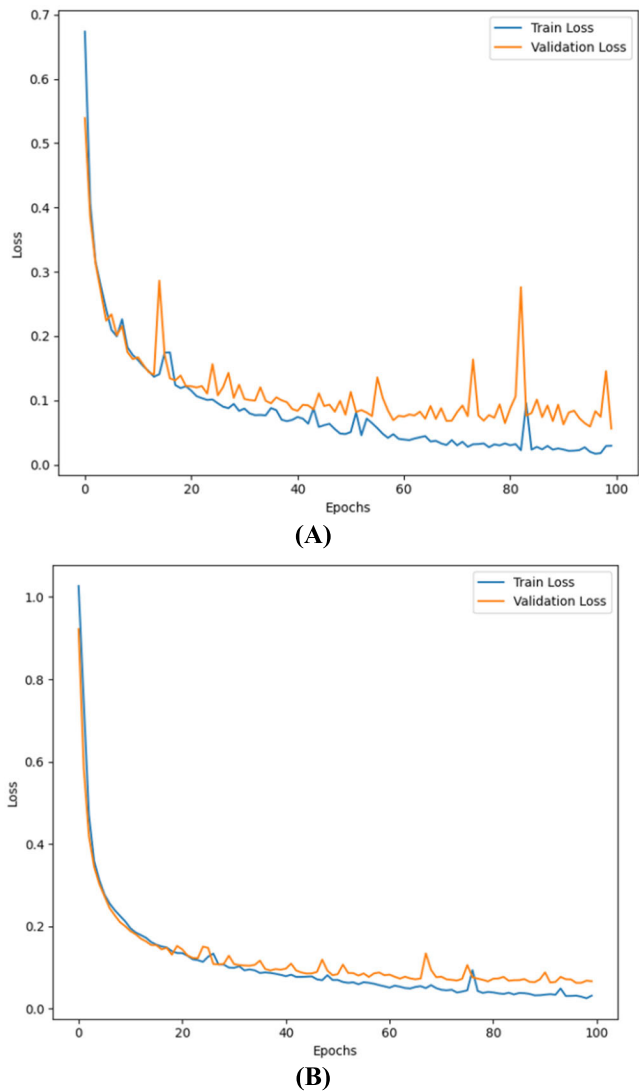


FIGURE 6. Train vs Val loss for German Credit dataset (A) Before weighted (B) After weighted loss.

TABLE 5. Classification metrics comparison before and after applying weighted loss on BISAID dataset.

Traditional TabTransformer with Cross Entropy Loss				Our's (Weighted)		
Class	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.76	0.83	0.79	0.87	0.78	0.82
1	0.92	0.89	0.90	0.96	0.88	0.92
2	0.88	0.85	0.86	0.83	0.95	0.89
3	0.82	0.84	0.83	0.86	0.95	0.90
Overall Accuracy			86.35%	Overall Accuracy		
				89.27%		

class, however, improves modestly from 0.88 to 0.90. These results indicate that the proposed weighted transformer effectively enhances sensitivity to minority classes while maintaining strong performance for majority classes.

Table 6 presents a detailed comparison of model performance using a classification report format. Initially, the traditional TabTransformer model, integrated with a standard binary entropy loss function, achieved a relatively high accu-

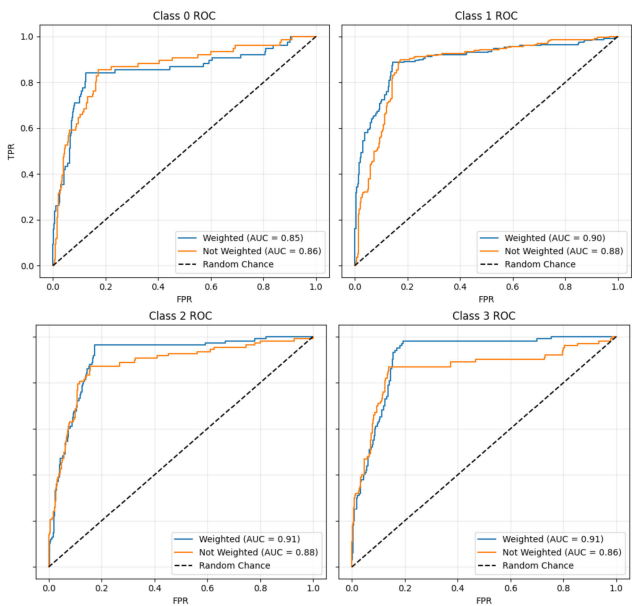


FIGURE 7. ROC curves comparing weighted and non-weighted multiclass model performance on BISAID database.

racy of 93%. However, the minority class (Class 0) suffered from low precision, resulting in a reduced F1-score, which indicates room for improvement in handling imbalanced datasets. Our research incorporates the TabTransformer model with a weighted binary entropy loss function, which has demonstrated significant improvements in performance. Not only did the overall accuracy increase from 93% to 95%, but all metrics across both classes also improved. Specifically, precision, recall, and F1-score for the minority class (Class 0) rose from 0.85, 0.93, and 0.89 to 0.88, 0.97, and 0.92, respectively. Importantly, these enhancements were achieved without sacrificing the precision or recall of the majority class (Class 1), which improved from an F1-score of 0.95 to 0.96. These findings illustrate that the weighted loss function is an effective solution for addressing class imbalances in binary classification tasks. By ensuring fair improvements across all metrics without degrading majority class performance.

TABLE 6. Classification metrics comparison before and after applying weighted loss on German Credit dataset.

Traditional TabTransformer with Binary Entropy Loss				Our's (Weighted)		
Class	Precision	Recall	F1-score	Precision	Recall	F1-score
0	0.85	0.93	0.89	0.88	0.97	0.92
1	0.97	0.93	0.95	0.99	0.94	0.96
Overall Accuracy			93%	Overall Accuracy		
				95%		

Figure 8 demonstrates a similar pattern to the ROC curve presented in Figure 7, where both figures illustrate the model's discriminative ability between positive and negative classes. The ROC curve in Figure 8 shows consistent improvements in sensitivity and specificity compared to the

baseline. In terms of AUC, our proposed technique incorporating weighted loss achieves an AUC of 0.96, compared to 0.91 in the baseline model. This improvement highlights the effectiveness of the proposed method in distinguishing between classes, leading to a more reliable and unbiased prediction model in imbalanced class scenarios without altering the distribution of the class.

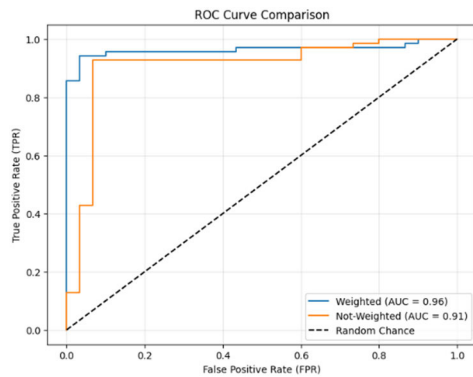


FIGURE 8. ROC curves comparing weighted and non-weighted multiclass model performance on German Credit database.

D. XAI POST-MODELING

After achieving the desired accuracy and mitigating model bias toward the majority class, the inference results become fair for both majority and minority classes. In this research, explainable AI (XAI) methods, specifically SHAP (SHapley Additive exPlanations), were employed to ensure that feature importance analysis remains unbiased. SHAP not only highlights the strong features associated with the majority class but also fairly represents the minority class, making the method valid and trustworthy for evaluating feature contributions across all classes.

As shown in Figure 9, the feature Financing Needs has the highest SHAP value, averaging around 0.65, indicating it is the most significant predictor in the model. Following this, Annual Sales and Total Assets contribute considerably, with average SHAP values of approximately 0.2 and 0.15, respectively. Features such as City and Labour have moderate impacts, with SHAP values near 0.1 and 0.05, respectively. Meanwhile, Province and Sector show the least contribution, with SHAP values below 0.05. This analysis demonstrates that the model primarily relies on financial and operational metrics, such as financing needs and annual sales, to make predictions. The lower contributions of features like province and sector suggest that regional and industry-based factors have less influence in this context.

Figure 10 shows the SHAP-based feature importance analysis for the German Credit dataset. Credit Amount is the most influential feature, with an average SHAP value of 0.12, followed by Age (years) and Status of Existing Credit, with SHAP values of 0.09 and 0.08, respectively. Other significant features include Duration (months) and Credit History (SHAP values around 0.07 and 0.05), while Savings

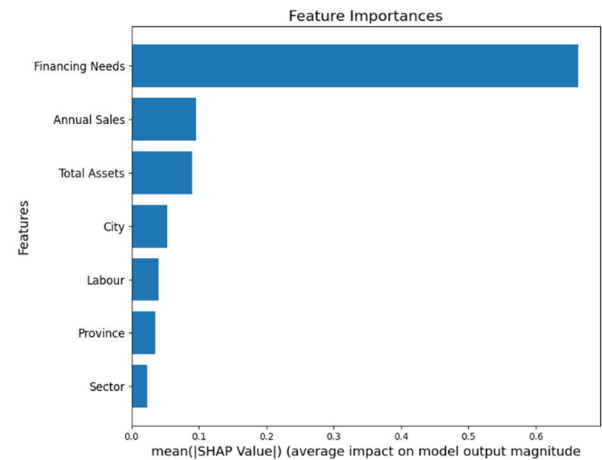


FIGURE 9. Feature importance analysis with SHAP on BISAID dataset.

Account/Bonds and Present Employment Status contribute moderately. Less impactful features, such as Foreign Worker and Telephone, have SHAP values below 0.02. This analysis highlights the importance of financial and demographic factors in credit risk prediction and demonstrates the interpretability of the transformer-based model using SHAP.

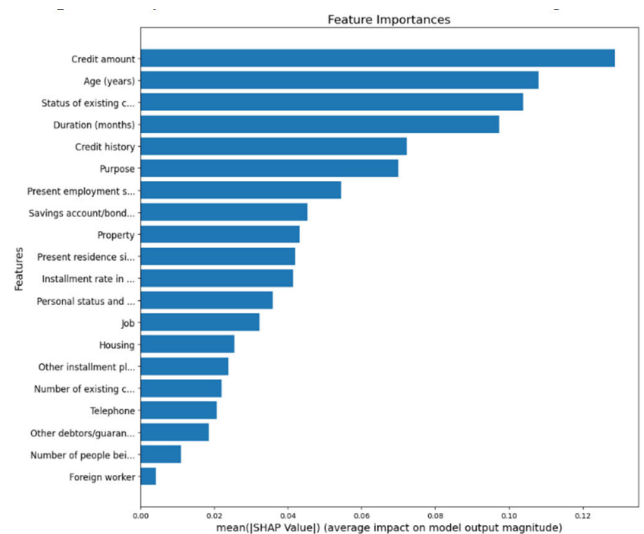


FIGURE 10. Feature importance analysis with SHAP on German Credit dataset.

This chapter emphasized the vital role of Explainable Artificial Intelligence (XAI) in ensuring interpretability and fairness in credit scoring models. Using SHAP (SHapley Additive exPlanations), key features such as “Financing Needs” in the BISAID dataset and “Credit Amount” in the German Credit dataset were identified, ensuring transparency in predictions. Weighted loss techniques addressed class imbalances, enabling fair representation of minority classes. The integration of XAI with the TabTransformer model provided a robust framework for accurate and equitable credit

evaluations, highlighting the importance of interpretable AI in promoting fairness and trust in financial decision-making.

V. CONCLUSION

This research introduces a novel framework for credit risk assessment, integrating weighted loss techniques with the TabTransformer model to effectively address class imbalance and enhance predictive accuracy. The weighted loss approach, applied to imbalanced datasets such as BISAID and German Credit, demonstrated significant improvements in performance metrics. For the BISAID dataset, the application of weighted loss increased overall accuracy from 86.35% to 89.27%, with AUC for minority classes improving from 0.88 to 0.91. Similarly, in the German Credit dataset, accuracy improved from 93% to 95%, and minority class precision and recall rose from 0.85 and 0.93 to 0.88 and 0.97, respectively. These enhancements ensured fairer representation of minority classes while maintaining robust performance for majority classes.

Building on this foundation, Explainable Artificial Intelligence (XAI) techniques, specifically SHAP (SHapley Additive exPlanations), were incorporated to avoid bias toward the majority class and make predictions more transparent for borrowers and stakeholders. SHAP enabled the identification of key features such as “Financing Needs” in the BISAID dataset and “Credit Amount” in the German Credit dataset, providing clear and interpretable insights into the factors driving credit decisions. This dual focus on fairness and transparency addresses the “black box” nature of traditional machine learning models, fostering trust and accountability in automated credit evaluation systems.

This research opens pathways for further advancements in AI-driven credit risk assessment. Future work can focus on scaling to larger datasets, incorporating real-time financial data, and integrating alternative credit sources. Adaptive weighted loss mechanisms could further enhance model performance. Collaboration with financial institutions will be key for real-world validation, ensuring compliance with regulatory standards while promoting fairness and transparency in automated credit evaluations.

ACKNOWLEDGMENT

This research is partially supported by the Vice-Rector of Research, Innovation, and Entrepreneurship at Satya Wacana Christian University through the Article Writing Bootcamp program and internal research grant.

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